Assignment 3: Github

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Assignment 3: Collaborating in Github

Introduction

The following assignment was conducted collaboratively in Github and involves analysing a dataset saved within the GitHub repository URL: https://github.com/harsha2412/bhds2010.git

The dataset TextMessages.cvs contains information about the number of texts sent by members of two groups (1 and 2) across two time-points (baseline and six months). Our goal is to create and interpret stratified boxplots and bar charts of text messages by group and time, as well as construct a table capturing summary statistics for the two groups by time. The README.md file within the GitHub was updated to reflect each collaborator's contributions to the project and the branching/pull request workflow. The working directory was set to the bhds2010 folder within GitHub.

Installing and Loading Packages

```
#Installing and loading in relevant packages

#install.packages("pastecs")
#install.packages("tidyverse")
#install.packages("knitr")
#install.packages("car")
#install.packages("Hmisc")
#install.packages("readxl")
library(readxl)
library(Hmisc)
library(pastecs)
library(tidyverse)
library(knitr)
library(car)
```

Data Preparation

```
#Reading in csv file from GitHub and assigning to data frame object named
#"text_wide" to distinguish wide format data.
text_wide <- read.csv("TextMessages.csv")
print("preview source file")
```

```
## [1] "preview source file"
```

```
head(text_wide, 10) # preview original data
```

```
##    Group Baseline Six_months Participant
```

```
## 1       1        52          32            1
## 2       1        68          48            2
## 3       1        85          62            3
## 4       1        47          16            4
## 5       1        73          63            5
## 6       1        57          53            6
## 7       1        63          59            7
## 8       1        50          58            8
## 9       1        66          59            9
## 10      1        60          57           10
```

```
class(text_wide$Group)
```

```
## [1] "integer"
```

```
#The group variable is currently numeric so we used factor() to convert to
#categorical
text_wide$Group <-factor(text_wide$Group,levels=c(1,2))

#Our visualizations (boxplot, bar chart) and summary statistics will utilize
#data in long format. We used pivot_longer from tidyr to reshape the data into
#long format, which will convert the two baseline and six months columns into
#one column capturing the time-point.
text_long <- pivot_longer(
  text_wide,
  cols= c("Baseline","Six_months"),
  names_to = "Timepoint",
  values_to = "Text_count"
  )
#Checking the class of the newly created column Timepoint
class(text_long$Timepoint)
```

```
## [1] "character"
```

```
#Since we want R to treat it as a categorical variable instead of a character
#variable, we again convert using factor() and order the two categories so
#baseline appears before six months
text_long$Timepoint<-factor(text_long$Timepoint,levels = c("Baseline","Six_months"))

#Moving participant number to the front of the data
text_long <- text_long |> relocate("Participant")
print("preview long file")
```

```
## [1] "preview long file"
```

```
head(text_long)
```

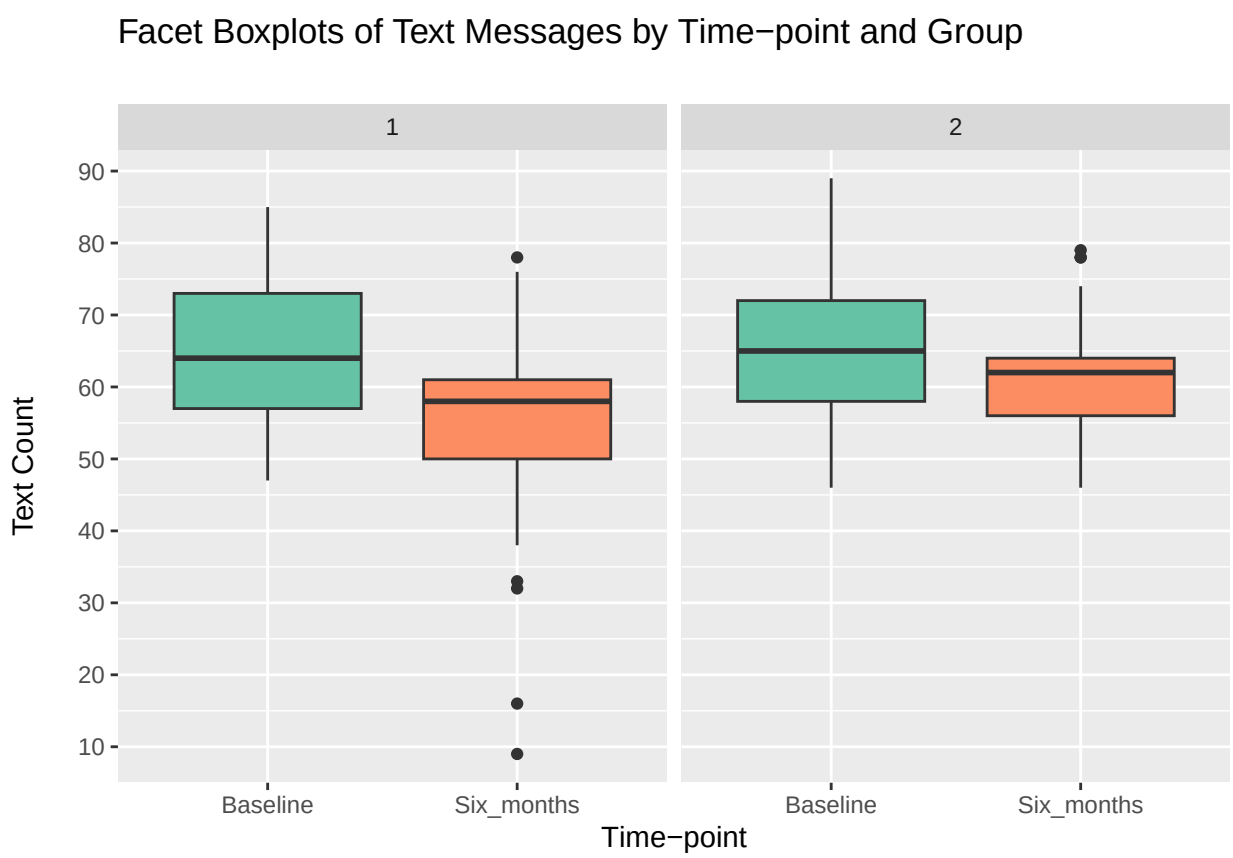
```
## # A tibble: 6 x 4
##   Participant Group Timepoint  Text_count
##         <int> <fct> <fct>           <int>
## 1           1 1     Baseline           52
## 2           1 1     Six_months         32
## 3           2 1     Baseline           68
## 4           2 1     Six_months         48
## 5           3 1     Baseline           85
## 6           3 1     Six_months         62
```

Visualization 1: Faceted Boxplot of text messages by Group and Time

```
#Creating faceted boxplots of text messages for each group and across the
#time-points using ggplot() and geom(boxplot). We color-coded by time-points and
#used facet_wrap(~Group) to display the plots side by side.

visualization1<- ggplot(
  data=text_long,
  aes(y=Text_count,
      x=Timepoint,
      fill=Timepoint))+
  geom_boxplot(show.legend=FALSE)+
  facet_wrap(~Group)+
  scale_y_continuous(breaks=seq(from=0, to=100, by=10))+
  labs(title = "Facet Boxplots of Text Messages by Time-point and Group\n",
       y="Text Count\n",
       x="Time-point")+
  scale_fill_brewer(palette = "Set2")

print(visualization1)
```



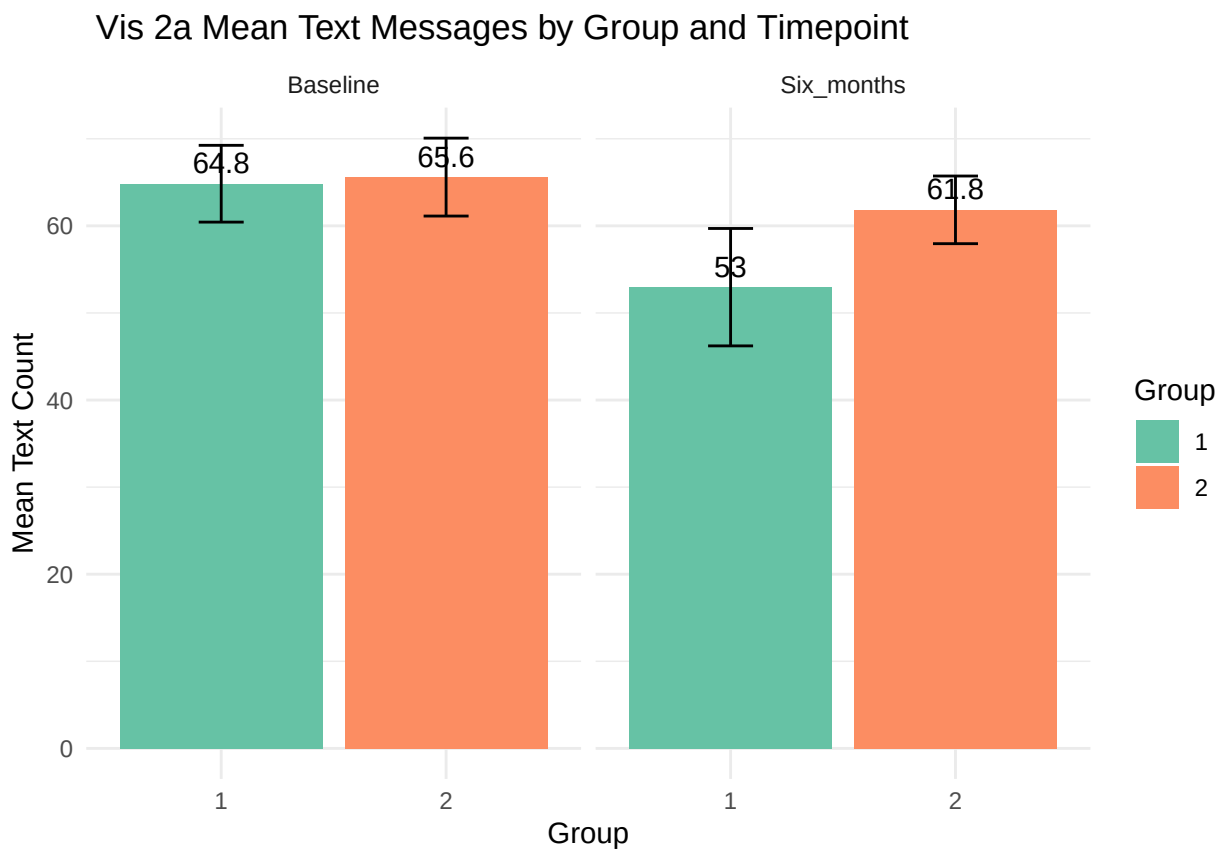

```
#We can compare the distributions of text messages across time-points (Baseline,
#6-months) for both groups (1,2) side by side. At baseline, both groups show
#similar distributions of text message counts, with approximately the same
#median values of 64/65. The interquartile ranges and overall spread also looks
#comparable, which suggests that the two groups had similar baseline
```

```
#characteristics.

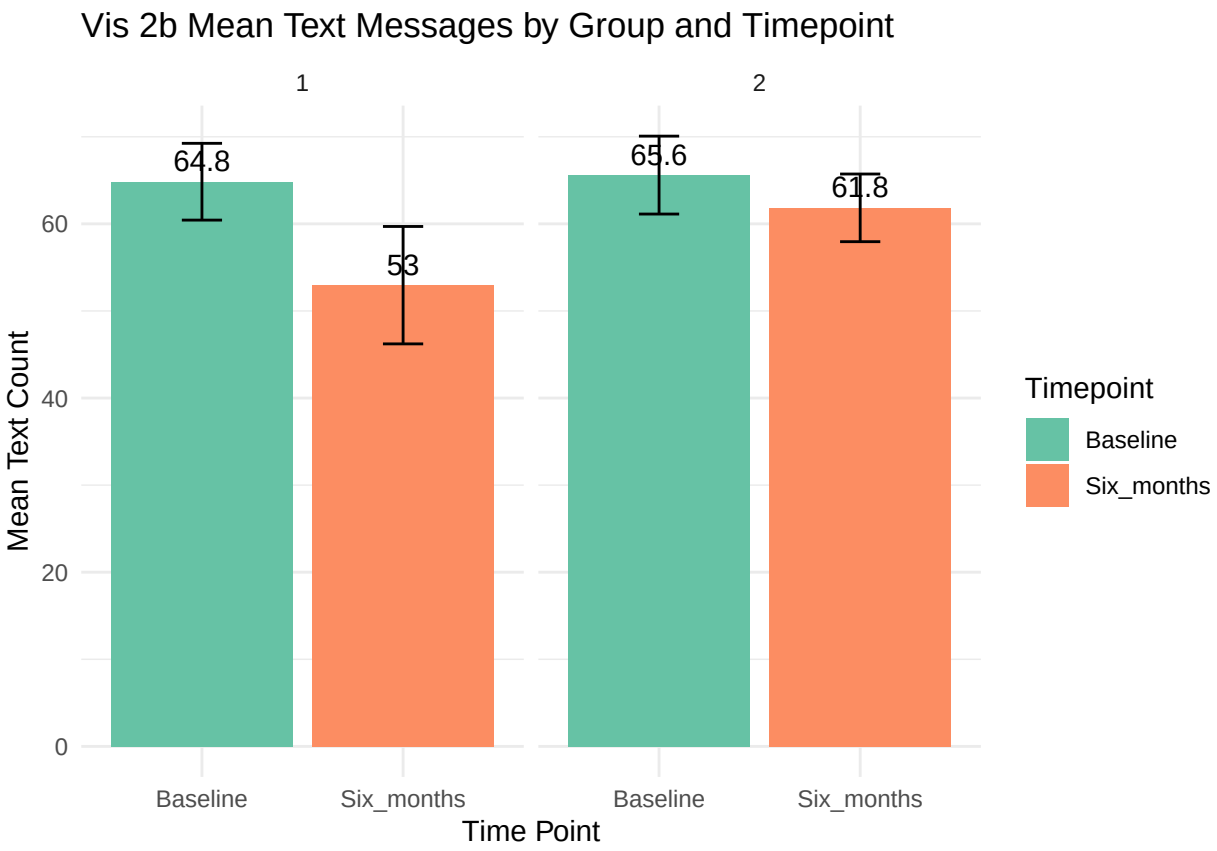
#At 6 months, we can see a decrease in median text message counts for both
#groups, though the reduction looks to be more for group 1 (from approximately
#64 to 58) when compared to group 2 (approximately 65 to 62). There are also
#noticeable differences in variability between the groups at six months. Group 1
#has a wider distribution, with several low outliers with values below 20
#messages as well as higher counts above 75. This suggests greater variability
#in text counts among participants in Group 1. In contrast, Group 2 has a
#narrower distribution with relatively higher messaging counts, suggesting more
#consistent messaging patterns across the participants.
```

Visualization 2: Faceted Bar Chart of text messages by Group and Time

```
# 2a creating the bar blots with facets. group on the x axis and timepoint as
# facets
  ggplot(text_long, aes(x = Group, y = Text_count, fill = Group)) +
    stat_summary(fun = mean, geom = "bar") + # using a 95% CI on the standard
  # error for the error capture.
    stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2) +
  # printing the mean on the plot for easier conclusions
    stat_summary(fun = mean,
                 geom = "text",
                 aes(label = round(after_stat(y), 1)),
                 vjust= -0.5) +
    facet_wrap(~ Timepoint) + # add facet for each group
    labs(
      title = " Vis 2a Mean Text Messages by Group and Timepoint",
      x = "Group",
      y = "Mean Text Count"
    ) +
    scale_fill_brewer(palette = "Set2") + # match Manya's color theme +
    theme_minimal()
```




```
## version 2 with timepoint on the x axis and the group as a facet
  ggplot(text_long, aes(x = Timepoint, y = Text_count, fill = Timepoint)) +
    stat_summary(fun = mean, geom = "bar") + # using a 95% CI on the standard
    # error for the error capture.
    stat_summary(fun.data = mean_cl_normal, geom = "errorbar", width = 0.2) +
    # printing the mean on the #plot for easier conclusions
    stat_summary(fun = mean,
                 geom = "text",
                 aes(label = round(after_stat(y), 1)),
                 vjust= -0.5) +
    facet_wrap(~ Group) + # add facet for each timepoint
    labs(
      title = "Vis 2b Mean Text Messages by Group and Timepoint",
      x = "Time Point",
      y = "Mean Text Count"
    ) +
    scale_fill_brewer(palette = "Set2") + # match Manya's color theme
    theme_minimal()
```



Summary Statistics of Text Messages by Group and Time

```
#We are first grouping the data by the group number and time-point, then
#computing the summary statistics that we want such as the mean, median, SD,
#range, etc., using dplyr from tidyverse.

sum_stat<- text_long |>
  group_by(Group,Timepoint) |>
  summarise(
  N=n(),
  Mean=mean(Text_count),
  SD=sd(Text_count),
  Median=median(Text_count),
  Min=min(Text_count),
  Max=max(Text_count),
  IQR=IQR(Text_count),
  .groups = "drop_last"
)

#To output a clean table of the summary statistics, we are using the kable
#function.

kable(sum_stat,
  caption="Summary Statistics of Text Messages by Group and Time-point",
  digits=2,
```

```
  align=c("c","l","c","c","c","c","c","c","c")
)
```

Table 1: Summary Statistics of Text Messages by Group and Timepoint

Group	Timepoint	N	Mean	SD	Median	Min	Max	IQR
1	Baseline	25	64.84	10.68	64	47	85	16
1	Six_months	25	52.96	16.33	58	9	78	11
2	Baseline	25	65.60	10.84	65	46	89	14
2	Six_months	25	61.84	9.41	62	46	79	8

```
#The summary table supports the trends observed from the previous plots. At
#baseline, the two groups had relatively similar mean numbers of text messages.
#Group 1 had a mean of 64.84 (SD = 10.68), while Group 2 had a slightly higher
#mean of 65.60 (SD = 10.84). The similar means and standard deviations confirm
#that the groups had comparable text messaging patterns at the start of the
#study.

#At six months, both groups experienced a decrease in the mean number of text
#messages as expected. However, the magnitude of the decrease differed between
#groups. Group 1 had a mean of 52.96 (SD = 16.33), representing an average
#decrease of 11.88 messages from baseline. Group 2 had a smaller decrease, with
#a mean of 61.84 (SD = 9.41), representing a reduction of 3.76 messages.

#When comparing means to medians, most of the subgroups have similar values
#hinting at symmetrical distributions. We only see a major difference in the six
#month time-point for group 1 where the median was higher than the mean,
#suggesting a left skew to the distribution. There were also differences in the
#variability of text messaging patterns at six months that were seen on the
#various plots. Group 1 demonstrated greater variability (SD = 16.33) compared
#to baseline, suggesting that participants' messaging behaviors became more
#dispersed. This is also reflected in the wider range of values (9-78). In
#contrast, Group 2 demonstrated less variability at six months (SD = 9.41), with
#a narrower range (46-79), confirming the more consistent messaging patterns
#among participants.
```

Interpretation and conclusions

Any concluding comments in addition to individual interpretations above

Visualization 1 conclusions

Overall, the boxplots suggest a reduction in text messaging frequency in both groups over the six-month period, with a more noticeable shift in Group 1. Additionally, Group 1 demonstrates greater variability in messaging behavior at the six-month time-point, whereas Group 2 shows a more consistent distribution. The presence of several low outliers in Group 1 suggests that changes in messaging frequency may vary considerably among individual participants.

Visualization 2 Conclusions

1. Text count for group 1 has dropped noticeably in the six months later period with wider error bars, suggesting more variability in response in the six month period. This is clearer in Vis 2b. Vis 2a shows

that the two groups were very similar in the baseline timepoint,

2. Text count for group 1 has much less pronounced difference between the two periods (65 to 62)

Summary table conclusions

Both groups demonstrated a reduction in text messaging frequency over the six-month period, but the decrease was more pronounced in Group 1 (11.88) compared to Group 2 (3.76). While the groups had comparable baseline averages(64.84 and SD = 10.68 for Group 1, 65.60 and SD = 10.84 for Group 2), their six-month averages diverged, with Group 1 (52.96 and SD = 16.33) reporting fewer messages on average than Group 2 (61.84 and SD = 9.41).

These descriptive findings support the interpretations provided by the boxplots and bar charts, and suggest a potentially greater change in messaging behavior in Group 1. However, descriptive statistics alone cannot establish whether the observed differences are statistically significant or attributable to group assignment. Further inferential analysis, such as comparing within-group changes and between-group differences in change, would be necessary to determine whether the differences represent meaningful intervention effects.

One statistical consideration: Because the same participants were measured at baseline and six months, a paired analysis would be appropriate for evaluating changes within each group.